Machine Vision and Applications (2011) 22:389–407
DOI 10.1007/s00138-010-0246-3

ORIGINAL PAPER

A cognitive and video-based approach for multinational License Plate Recognition

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Received: 9 May 2009 / Revised: 4 January 2010 / Accepted: 10 January 2010 / Published online: 1 March 2010

Abstract License Plate Recognition (LPR) is mainly regarded as a solved problem. However, robust solutions able to face real-world scenarios still need to be proposed. Country-specific systems are mostly, designed, which can (artificially) reach high-level recognition rates. This option, however, strictly limits their applicability. In this paper, we propose an approach that can deal with various national plates. There are three main areas of novelty. First, the Optical Character Recognition (OCR) is managed by a hybrid strategy, combining statistical and structural algorithms. Secondly, an efficient probabilistic edit distance is proposed for providing an explicit video-based LPR. Last but not least, cognitive loops are introduced at critical stages of the algorithm. These feedback steps take advantage of the context modeling to increase the overall system performances, and overcome the inextricable parameter settings of the low-level processing. The system performances have been tested in more than 1200 static images with difficult illumination conditions and complex backgrounds, as well as in six different videos containing 525 moving vehicles. The evaluations prove our system to be very competitive among the non-country specific approaches.

Keywords Multi-country LPR · Cognitive loops · Probabilistic Levenstein distance · Hierarchical Neural Network · Reeb graph

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1 Introduction

License Plate Recognition (LPR) is an important research area that has been widely studied in the last two decades. It includes many promising applications such as unattended parking lots, security control of restricted access areas, traffic law enforcement, congestion pricing, automatic toll collection, speed limit enforcement, identification of stolen cars, etc. A lot of industrial Car License Plate Recognition (CLPR) systems emerged since the 1980s, and today commercial OCRs are available at very affordable prices and impressive recognition rates (98–99%). Thus, LPR is widely regarded to be a solved problem. However, most of existing commercial solutions are only applicable to controlled conditions (illumination, viewpoint, etc.), or require high-resolution and sometimes specialized imaging hardware possibly combined with infrared strobe lights. Moreover, many approaches optimize their algorithms for a specific kind of national plates.

Thus, providing a robust LPR system that can deal with various LP types while managing difficult conditions and achieving real-time remains a very challenging task. In this paper, we address this problem with a cognitive approach. Contrary to recent works [19,30,40], we do not try to design the low-level steps to adapt to various LP types. We rather tackle the multi-national LPR by using cognitive loops, so that the high level analysis can automatically detect low-level step failures, and eventually adapt their parameters to compensate for them.

The remainder of the paper is organized as follows. Section 2 proposes a state of the art of the existing LPR approaches. Section 3 details the low-level steps of the system. Then, our three main contributions are explained in Sects. 4, 5 and 6, respectively. Results illustrating the approach efficiency and comparing the system performance with other methods are presented in Sect. 7. Finally Sect. 8 concludes the paper and proposes directions for future works.

2 State of the art

In this section, we propose a general overview of the LPR computer vision based approaches, classified into LP detection (Sect. 2.1), LP recognition (Sect. 2.2), and video-based systems (Sect. 2.3). We outline the use of national *priors* on LP format in Sect. 2.4.

2.1 Plate detection

Basically, we can classify LP detection approaches into two groups: appearance-based and gradient-based methods. The former class relies on color or textural features to discriminate the plate from the background. The algorithms using color exploit the particular a priori known color of a given national LP [8,46,47]. These approaches are clearly country-specific. Moreover, they are not robust to illumination changes, and the segmentation accuracy directly depends on the discriminability between the given color and the background. Textural features can then be employed alone or in combination with color cues to improve the detection accuracy [1,22,32], using Gabor filters, wavelets, *etc*. These approaches are most of the time coupled with machine learning tools in order to identify features that best discriminate LPs from the background. For example in [32], the authors propose to combine a texture and yellow detector to localize Dutch LPs, using a fuzzy logic strategy for the classification. Kim et al. [22] segment the license plate using SVM in conjunction with the CamShift algorithm, finding a compromise between accuracy and time processing. The appearance-based approaches proved to be efficient by the rich features extracted and the training step involved. However, the efficiency of the detection is always related to the training dataset, and it is not clear how well the detector will generalize in other situations, particularly with different types of national LPs.

On the other hand, gradient-based approaches model the LP as a high contrast image area. Thus, some authors propose to use the local variance magnitude as a feature for locating LPs [17], since other approaches use mathematical morphology operators as a post-processing step [18]. Another possibility to identify high contrast areas consists in using the Hough Transform for detecting lines [12]. An important subclass of the gradient-based approaches belong to the document analysis community, and corresponds to text detection. In the LPR context, Cui et al. [9] compute the spatial variance on each row of the image, making a strong variance value interact with the presence of text. If the text is coarsely vertically oriented in the image, this assumption being actually fulfilled for LPR applications, a commonly used technique consists in computing the horizontal and vertical accumulation of vertical gradients [25]. The previous methods can be considered as bottom-up approaches, where an edge map is computed and processed to localize the LP. Other more top-down approaches consist in using sliding-window pattern-recognition detectors. For example, the AdaBoost algorithm was used in [7,10] for LP detection on low resolution video frames, using an enriched set of Harr features. Gradient-based approaches rely on edges features. This make them better candidates for being applicable in real applications, with unknown LP national formats. On the other hand, they model LPs less accurately, and post-processing steps able to remove false positive inevitably have to be carried out to achieve good performances in real situations.

2.2 Plate recognition

Once LP is detected, one possibility for the recognition consists in performing character segmentation and recognition as a whole, using template matching techniques [8,10]. These top-down strategies estimate the likelihood of a given template at a given location, mostly by computing a (normalized) cross-correlation measurement. These methods offer the advantage of ignoring the arduous individual characters extraction step. They are, however, very time consuming even under the strong assumptions of no tilted plates or fixed-sized characters. In addition, these approaches are not learning-based, which clearly limits their generality.

2.2.1 Character extraction

Thus, the majority of the approaches use more bottom-up strategies, first attempting at locating the different characters. In that perspective, the image plate binarization is a common pre-processing step. Connected component labeling [31,32] has intensively been used for the character extraction purpose. This strategy is accurate but is, however, extremely sensitive to noise, a single mis-labeled pixel potentially leading to segmentation failure. Alternatively, histograms projection (HP) technique [11,12,23] consists in searching for changes from valleys to peaks, by counting the number of black pixels per column in the projection. This strategy is more robust to noise, but also less accurate. In addition, since it is easily affected by the rotation of LP, segmenting the skewed LP becomes a key issue to be solved. Another strategies rely on global optimization schemes, i.e. the goal becomes to obtain a compromise between character's spatial arrangement and single character's segmentation result. Hidden Markov chains have been used to formulate the dynamic segmentation of characters in LP [14]. Global optimization schemes increase robustness by explicitly modeling the LP format. This modeling must remain sufficiently broad to be efficient in a multinational context.

Recently, many approaches try to combine different strategies to improve the segmentation performances. Chang et al. [6] choose to extract connected components first, and to apply a set of three operators (delete, merge, and split) to determine if a contour satisfies the LP structural constraints. Nomura et al. [33] propose an adaptive approach for managing seriously degraded characters in low resolution images. Based on a HP technique, the algorithm is able to detect fragmented, overlapped and connected characters, and to properly segment them.

2.2.2 *Optical Character Recognition*

The Optical Character Recognition (OCR) is the heart of the LPR system. It traditionally consists in two main steps: feature extraction and feature classification. Regarding methodology, we can distinguish two kinds of strategies: statistical and structural approaches.

In statistical approaches, the input pattern is composed of vectorial data, and the classification stage can be seen as a statistical decision process. For OCR applications, the most basic feature that can be considered corresponds to the overall image extracted. Most approaches incorporate it into their LPR system [1–3,11,23,31]. However, as illustrated in Fig. 1a, this strategy is not very robust when dealing with real images, because the input vector has very limited invariance properties (it is even non translation invariant). Indeed, the two binary images have input vectors (composed of the sequence of image pixels) that significantly differ, although they belong to the same class. Nijhuis et al. [32] reported a long time ago that extensive experiments showed that it is rather impossible for bit wise image input to obtain an error rate below 1% with low resolution images. Thus, semantically richer representations have been studied to manage real characters. A general review of shape descriptors is beyond the scope of the paper and the reader can refer to [49]. Regarding OCR applications, the following descriptors have been proposed: Fourier Descriptors [41], Curvature Scale Space [24], Zernike moments [4], Hu's moments [5]. Gabor features [13,20,28] Regarding classification, it can be performed using template matching [31], Hausdorf distance [29], neural networks [1,11,32] including Kohonen self-organized feature maps [6], Hidden Markov Models [12], SVM [23], etc.

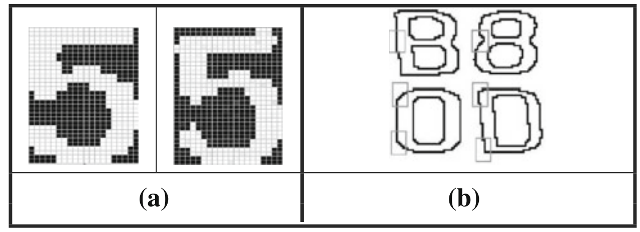

Fig. 1 Character recognition: invariance and locality. **a** Pixels are not a robust input feature. **b** Similar characters robust input feature only differ locally, from [6]

For shape recognition applications, the strength of statistical methods is their robustness to noise. However, some small differences between similar patterns may be difficult to distinguish. As pointed out in [6,35] for character recognition purposes, statistical approaches are thus efficient to discriminate the majority of the classes, but largely decrease with "similar" characters (such as '8'/'B' or '2'/'Z' and so on), particularly as the image quality is low. This is illustrated in Fig. 1b. On the other hand, structural approaches decompose the patterns into simpler primitives and obtain the properties of primitives and/or their relationships. Structural methods are closer to human recognition strategy [36], and are more powerful to recognize similar patterns.

2.3 Video-based approaches

Despite the huge amount of academic study on the LPR topic, few video-based works explicitly use the temporal information of the sequence. In most systems, either the acquisition conditions provide static images only, or the image sequence is triggered by another device and only some frames are captured and analyzed independently. However, capitalizing on the temporal aspect of the video can highly increase system performance. Basically, using the temporal information consists in tracking vehicles over time to estimate LP motions and make the recognition step more efficient. There are two different kinds of strategy to achieve that goal. One possibility consists in using the tracking output to obtain a single high resolution image from multiple, subpixel shifted, and noisy low-resolution observations. This technique is known as super-resolution [9,42]. For example, Dlagnekov and belongie [10] propose to detect the license plate using an Adaboost classifier, and to track it using a data-association approach, before performing the OCR on the sub-sampled image. Alternatively to super-resolution techniques, merging the high-level outputs of the recognition to take a final decision can be attempted. For example, Arth et al. [2] propose to combine the classification results of subsequent frames, using the history provided by the tracking module. However, a simple majority voting for each position within the character string is applied, selecting the character achieving the highest count. This approach will not be very robust to a variable length of the string over time, making the success of the temporal averaging dramatically depends on the efficiency of the character segmentation at each time step.

2.4 Country-specific and multinational Systems

A crucial difference among LPR systems is related to their robustness to LP format variation. Indeed, it must be pointed out that most of the approaches incorporate a set of a priori knowledge about LP format [6,8,12,32,23,31]. Therefore, these LPR systems are optimized for recognizing a given specific LP format, and we refer to them as "country-specific" approaches. The *prior* knowledge can be used at various steps:

1. During LP detection, by using features that easily discriminate LP from the background, e.g. color, as detailed in Sect. 2.1.
2. During character extraction, by assuming a *prior* known LP structure: number of characters, distance between them, etc. As character extraction often constitutes the bottleneck of the overall LPR system, these national *priors* boost "country-specific" performances.
3. During character recognition, by using syntactic rules. For example, these syntactic rules can help to filter out false positives from character extraction, or to infer the character type (letter vs. digit, for example).

For example, Naito et al. [31] provide a system whose extraction and recognition is optimized for Japanese LPs, which are composed of a district office name, a classification number, a prefix, and registration numbers. To extract the registration number, the authors first isolate each of the other item, and make use of the model of the Japanese LP to make the segmentation efficient. Regarding recognition, the characters are known to be numbers, making the classification task much more easier. Thus, country-specific approaches can artificially reach excellent performances, but they will fail at recognizing an even slightly different LP format.

Although there is a huge literature in the general LPR topic, approaches relaxing LP format *prior* are relatively recent, and the amount of work is relatively limited [3,19,26,30,35,40]. In this paper, we denote these approaches as *multi-national* LPR systems. In addition, if the approaches being able to manage various LP formats regarding detection and extraction are now common, providing systems ignoring syntactical rules remains a very challenging task. Some recent works address this problem with different strategies.

First, some systems manage multi-national recognition by using very restrictive application conditions. For example, Lee et al. [26] propose a multi-national system, but applying an infrared sensor to trigger the image capture largely facilitates their approach. This heavy acquisition condition ignores most of the difficulties occurring with standard systems: scaling problem (LP is captured at a fixed size), illumination variations, character extraction, etc.

Other approaches explore strategies for learning variable character fonts. For example, Mecocci et al. propose in [30] an OCR method where Generative models(GM) are proposed to produce many synthetic characters whose statistical variability is equivalent to that showed by real samples. They validate their approach ability to improve the OCR performance. However, the method mainly concerns the OCR extensibility instead of the whole LPR method.

Some recent works formulate the multi-national recognition as a LP format estimation problem. In [40], the authors propose a LPR method in multi-national environment with character segmentation and format-independent recognition. The multinational recognition consists in firstly estimating the nationality among a set of predefined ones, then using the known format to improve the recognition. Although the approach is interesting, it is only evaluated with two (Israeli and Bulgarian) different nationalities. In addition, the OCR method is not based on learning, which clearly limits its extensibility. In [19], the authors propose to manage the multi-style LPR by representing the styles with four quantitative parameters (plate rotation angle, plate line number, character type and format). The approach is interesting, because the different parameters can model various LP styles, i.e. not only various LP countries but they can also model different acquisition condition (e.g. skewed LPs). The approach has been evaluated with five different nation LPs, and a thorough validation of the system is proposed. These recent works dealing with LP format estimation are the most relevant to us regarding application, since they constitute the most ambitious approaches towards multi-national LPR. However, regarding methodology, we tackle the problem with a different approach. Indeed, since they parametrize the low-level steps for managing various nationalities, we rather rely on a video-based strategy and on cognitive loops to make the recognition more efficient.

2.5 Proposed approach and contribution

In this paper, we propose a multi-national LPR system that can deal with various LP types while managing difficult conditions and achieving real-time. Our approach, illustrated in Fig. 2, is only restricted to LPs which characters lie on a single row, and dedicated to recognize alphanumeric symbols of the Latin alphabet.

To achieve this goal, our methodological contribution is three-fold. First, the OCR is made extremely efficient by the use of an approach combining statistical and structural

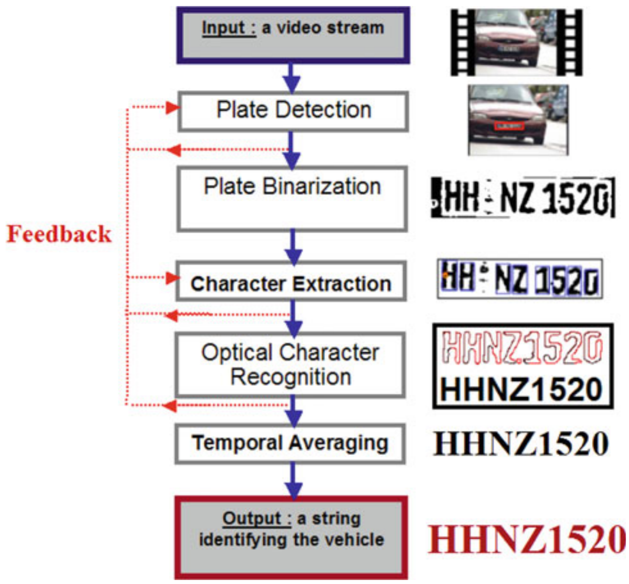

Fig. 2 Global system architecture

strengths. Secondly, we propose an efficient approach for combining the different OCR over time. This highly improves the recognition performances, by taking advantage of distinctive frames where the OCR can be easily achieved. Finally, cognitive loops provide a LPR system able to model the context in order to increase its performances. These feedback steps (shown with red dashed lines in Fig. 2) analyze the relevance of the algorithm at several critical stages. In case of failure detection, it leads to the adaptation of the parameters of the lower level steps.

3 Plate detection and character extraction

3.1 License plate localization

As we aim at managing various LP national formats, we choose a gradient-based strategy for LP detection. Each frame is thus processed to localize areas with a high vertical gradient density. After a pre-processing step (contour enhancement), we apply a vertical Sobel mask filter, and perform a gradient accumulation approach inspired from [25].

The filtered image is then thresholded, leading to the final labeling, where the pixel is identified as 'text' or 'non-text'. Connected component analysis is then carried out on the binary map, as illustrated in Fig. 3 a. A set of N candidates for the license plate is thus extracted, and sorted by decreasing order as follows:

$$S_G = w_1 \times S_{\text{Grad}} + w_2 \times S_{\text{Ratio}} + w_3 \times S_{\text{Shape}} + w_4 \times S_{\text{GTrack}} \quad (1)$$

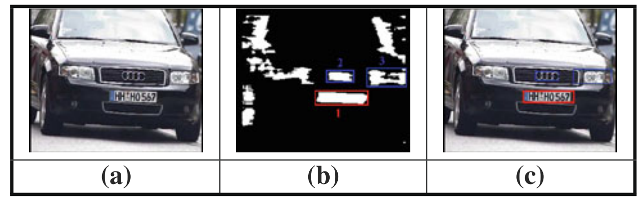

Fig. 3 Gradient-based license plate detection

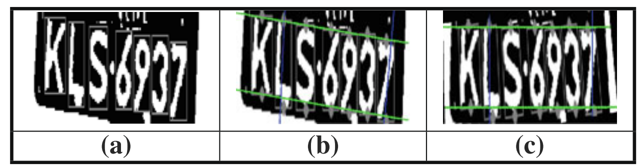

Fig. 4 Metric rectification of the image plane

S_G combines different priors including gradient, aspect ratio, shape and tracking criterion. The most probable candidate is processed, its successors are stored and may be considered in a feedback loop (see Sect. 6), as illustrated in Fig. 3c.

3.2 Plate binarization

The LP region being identified, a binarization step is then carried out to discriminate text pixels from background. We use a local version of the Fisher/Otsu algorithm [34], i.e. we estimate binarization parameters for a set of sliding windows, by tiling the LP candidate area. This makes it possible to be robust to illumination variations.

3.3 Character extraction

Starting from the binary LP image, we make use of a Connected Component Labeling (CCL), leading to a preliminary set of contours. To remove outliers, we use a simple geometrical reasoning on the individual connected components (Fig. 4a). In addition, we incorporate more global *priors* for modeling the different contour assembly, mainly relying in linearity, parallelism, and distance between characters. This is illustrated in Fig. 4b. Finally, we perform a metric rectification, that will enable us to preserve parallelism, orthogonality and aspect ratio between the plane of the license plate and the image plane [15] (see Fig. 4c).

3.4 Parameter setting: discussion

We recall that the low-level steps (i.e. LP detection, binarization and character extraction) are designed for providing a multinational system. In that context, the feature extraction must be country-invariant. In counterpart, the low-level steps performances are intrinsically limited with such invariant image descriptors. Therefore, we use the following strategy regarding parameter settings.

First, the parameters are learned by a cross-validation procedure, or even experimentally setup. Despite this first optimization stage, we experimentally notice that some false positives and false negatives remain, due to the high level of invariance required (invariance vs. specificity trade-off). It supports our intuition that a *context modeling* is required to improve efficiency and resolve ambiguities. Thus, depending on the higher level module analysis, the most critical parameters may iteratively be re-estimated by using cognitive loops (Sect. 6). For example, the connected component extraction scheme presented here is made more robust by the aspect ratio analysis of each contour, potentially leading to the conjoint use of the projection histogram technique. We strongly believe (and verify experimentally) that such a context modeling strategy is well-founded in the LPR context.

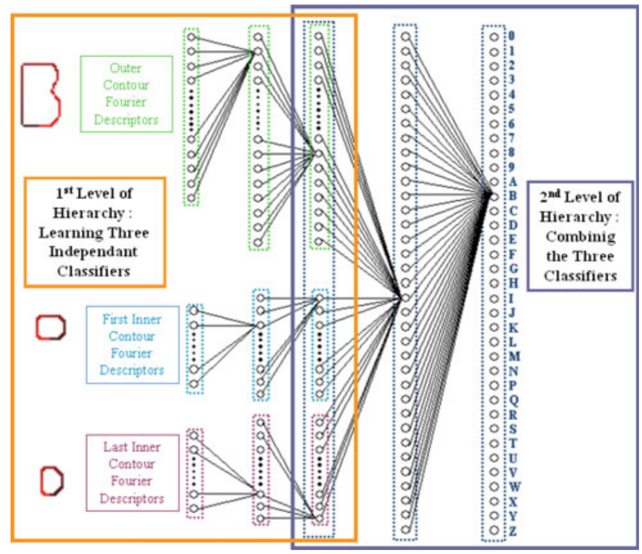

Fig. 5 Statistical classifier for OCR architecture

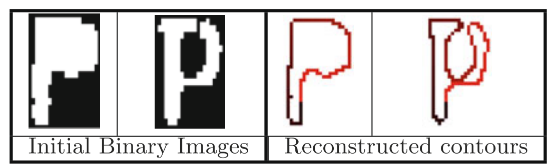

Fig. 6 Our HHN can efficiently learn characters with topological variations

4 Character recognition

The OCR is managed by an hybrid strategy, combining the advantages of statistical and structural methods. The approach can be sketched as follows: For each extracted character, a statistical recognition is carried out. This approach is extremely efficient for the majority of the classes, but the performances sensitively decrease for "similar characters", e.g. '8'/'B'. In that case, we adopt a strategy that revolves around decomposing the characters into structural elements.

4.1 The statistical classifier

The statistical part of the OCR consists in extracting Fourier Descriptors (FD) from the inner and outer contours of the image, and using a Hierarchical Neural Network (HNN) to perform the classification (see [43] for more details). FD provide a discriminative signature of the contour of an object [37]. In addition, since the general shape of the object is represented by the lower frequency descriptors, this subset provides a shape signature that is robust to noise. Thus, FD have been intensively used for shape matching applications, and prove to be superior to alternative representations, such as Autoregressive models [21], CSS or Zernike moments [48].

The architecture of the HNN is shown in Fig. 5. At the first level, three different perceptrons, that we denote Cl_1, Cl_2 and Cl_3, are used. They are intended to recognize the external contour and the two possibly existing internal ones. The second level of the hierarchy takes as input the results provided by the three previous neural networks and carries out a final recognition, resulting to an output vector quantifying the probability of each possible character.

In our LPR context, the hierarchical architecture brings us two major advantages. First, the shape information inherent to each inner/outer contour is wholly capitalized on. For example, despite '6' and 'Z' differ each other from the presence of an inner contour, we want our classifier to learn to discriminate them by means of the outer contour alone. Secondly, the hierarchy makes it possible to robustly learn characters with topological variations, as those illustrated in Fig. 6. These situations are actually common due to lower level step failures. As we experimentally validate, our HNN makes the learning stage more efficient than using a single-level (flat) classifier.

4.2 Structural analysis

As we discuss in Sect. 7, the HHN reaches excellent recognition rates in testing for the majority of the classes, i.e. more than 99%. The performances, however, sensitively decrease for similar classes, e.g. '8'/'B'. Indeed, FD are neither adapted to identify local details nor to capture prominent features as right angles. Thus, we present here a structural approach that is more adapted to differentiate similar characters.

We have chosen a structural approach based on discrete geometry tools to distinguish ambiguous characters in our system. We propose to describe the topology and the shape of such an example with an algorithm inspired from [44].

More exactly, this process performs the Reeb graph [38] of the binary object I and a simple polygonal structure inside I. Our approach is applicable to every classification schemes of ambiguous characters: '8'/'B', '2'/'Z', 'V'/'Y', etc. We first choose an order to treat the pixels of I, e.g. from left to right or from top to bottom. Then, our structural algorithm can be sketched as follows (see also Fig. 7):

1. Group pixels of I to build as large as possible irregular pixels (or *cells*). We denote those k cells $\{C_i\}_{i=1,k}$. This part could be compared to a Run Length Encoding scheme, where we have decided to merge together pixels with the same height. In 'B' and 'Z' examples, the cells should be respectively greater than in the '8' and '2' cases.
   While this incremental process is performed, we build our topological and geometrical representation:
2. In the Reeb graph G_I, the nodes link connected components of I together, these latter being maybe thick arcs of this object. Considering the order we have chosen before, a 'B' (with a left-to-right order) starts with one component, and is thus represented by one node in G_I (see Fig. 7). So, in general, a 'B' example should have less nodes than an '8' sample, which starts with two components (and two nodes in G_I). In the same way, a '2' example (with a top-to-bottom order) starts by one branch, divided into two parts, which leads to the construction of three nodes in the graph. Such an example has more nodes than a 'Z' image, where such a division does not exist.
3. The polygonal structure P_I is computed by tracing the longest line segments through the arcs of I, according to the graph G_I. There may be more points in P_I if I represents an '8' image than in a 'B' example. In the same way, P_I contains more points in a '2' image than in a 'Z' sample.

In our system, we have implemented a structural classifier based on AdaBoost algorithm [39], with feature vector V defined as follows:

$$V^T = \begin{pmatrix} \#\{p = (x, y) \in P_I\} \\ \max_{i=1,k}\left(S(C_i)\right) \\ \#\{n \in G_I\} \\ HNN(C_1) \\ HNN(C_2) \end{pmatrix} \qquad (2)$$

Where $S(P_i)$ is the surface of the pixel P_i. Those elements allow distinguishing in a general manner local shape features of ambiguous characters. Consider for example the left part of '8'/'B' images. The number of points in the polygonalization increases when this part is more rounded ('8' case). The largest cell in $\{C_i\}_{i=1,k}$ efficiently stands for a large straight part in the letter ('B' case). The number of Reeb graph nodes represents the number of arcs (two for the '8' case, and one for the 'B' case). As we show in the performance evaluation, the

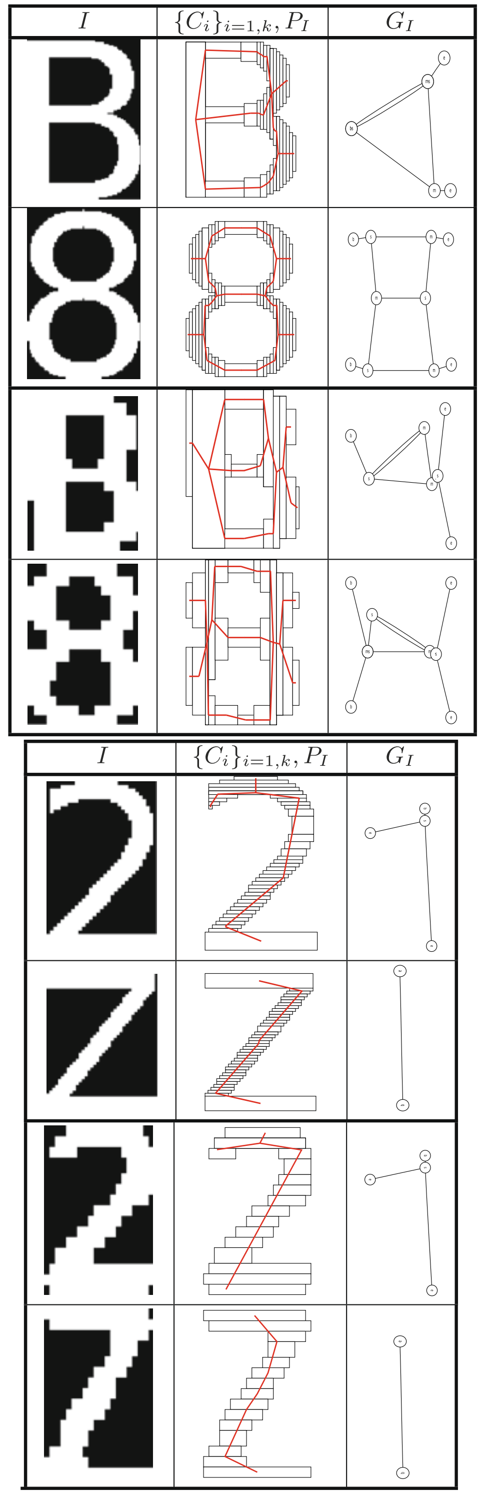

Fig. 7 The elements computed by the structural analysis The Reeb graph G_I and the polygonal structure P_I are depicted for "perfect" (*first and second rows*) and "real" (*third and fourth rows*) letters '8', 'B', '2' and 'Z'

recognition rate for the statistical classifier is slightly lower for discriminating similar images than for the other classes. However, it is largely higher than random. For that reason, the output of the Hierarchical Neural Network for the C_1 and the C_2 classes (that we denote $HNN(C_1)$ and $HNN(C_2)$, respectively) is inserted in the feature vector of the structural classifier, to provide a combined strategy.

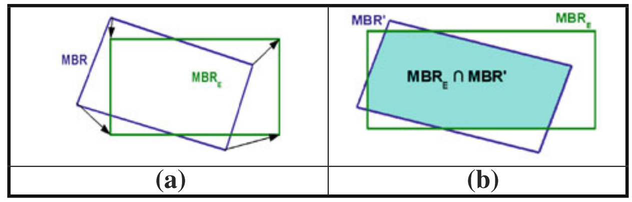

Fig. 8 License plate tracking: **a** estimated LP position and **b** $S_{\text{GTracking}}$

5 Making use of temporal coherence

In this section, we explain our approach for explicitly taking advantage of the temporal aspect of the video, and our goal is two-fold. First, we propose an approach for tracking an unspecified number of vehicles in the sequence (Sect. 5.1). During the tracking process, we update the recognition results to provide a mean string of the plate (Sect. 5.2). We propose here to perform the temporal averaging at the highest level of the algorithm, after the OCR is carried out. An alternative approach would consist in making use of super-resolution techniques [10,45]. However, we prefer here to average the OCR results at each time step. Indeed, it makes our algorithm able to use the high-level analysis to improve the low-level steps (see Sect. 6).

5.1 License plate tracking

We define a tracking score S_{Tracking} between a plate identified at time t and a new candidate being recognized at time $t+1$:

$$S_{\text{GTracking}} = w_G \times S_{\text{GTracking}} + w_R \times S_{\text{RTracking}} \tag{3}$$

w_G and w_R are parameters that have been learnt from training data, using a cross-validation procedure. $S_{\text{GTracking}}$ corresponds to a geometrical component that defines a similarity measurement involving position and speed:

$$S_{\text{GTracking}} = \frac{\text{Area}(MBR_E \cap MBR')}{\text{Area}(MBR_E \cup MBR')} \tag{4}$$

MBR' is the LP Minimal Bounding Rectangle (MBR) at time $t+1$ and MBR_E is the estimated position of the LP MBR from its previous location at time t. When a vehicle is being tracked, $S_{\text{GTracking}}$ is directly used for the LP detection (see Eq. 1 in Sect. 3.1).

The features tracked for estimating MBR_E correspond to the four corners of MBR. The location of each point in the image and its speed are used. A simple first order motion model is applied to predict the corner position at the next time step, as illustrated in Fig. 8a. When a new candidate for the license plate is detected in the subsequent frames, its minimal bounding rectangle MBR' is computed and matched against MBR_E using Eq. 4, as illustrated in Fig. 8b.

The second term of Eq. 3, $S_{\text{RTracking}}$, is related to the OCR result, and corresponds to a similarity score between the string extracted from the current plate and the previously updated buffer (see Sect. 5.2.1). Basically, it is based on the reasonable assumption that the current recognition result should be close to the previous ones when tracking the same plate.

The global tracking score S_{Tracking} defined in Eq. 3 is matched against a given threshold to define vehicles entries and outcomes. Thus, extending the algorithm for tracking multiple vehicles is straightforward. At each time step, we define a binary matrix that contains the tracking score S_{Tracking} between each detected plate and each previously tracked vehicle. Solving for best correspondences in this matrix makes it possible to track an unspecified number of vehicles.

5.2 Updating License Plate Recognitions over time

As a vehicle is being tracked, we take advantage of the temporal coherence by updating its LP over time. Let us denote C to be the string of N characters representing current LP detection, and B to be the previously updated buffer (M characters).

5.2.1 registration

The first issue to be solved corresponds to determine how to match $C[i]$ and $B[j]$, for each $\{i, j\} \in \{0; N\} \times \{0; M\}$. This is a registration problem, i.e. we have to find the best match for each of the character in $C[i]$ in C. To achieve this goal, our approach is based on the Levenstein distance [27], or Edit Distance (ED), to measure the similarity between strings.

As a toy example, let us consider that $B =$ ‘3368XB69’ and $C =$ ‘338869’ (two characters of the current string have not been extracted with respect to the buffer). Figure 9 illustrates the registration problem, and shows the result when using the Levenstein algorithm. The ED between the two strings is 3. During the backtracking step, there exists three competitive paths, showing that the Levenstein algorithm is ambiguous in this case.

In our context, we propose to adapt the ED by using the likelihood of the OCR, that approximates the a posteriori distribution of the different classes. Thus, during the backtracking phase of the algorithm, we compute the distance C_s

$$\begin{pmatrix} 3 & 3 & 6 & 8 & X & B & 6 & 9 \\ 3 & 3 & & 8 & 8 & & 6 & 9 \end{pmatrix} \quad \begin{pmatrix} 3 & 3 & 6 & 8 & X & B & 6 & 9 \\ 3 & 3 & 8 & 8 & & & 6 & 9 \end{pmatrix} \quad \begin{pmatrix} 3 & 3 & 6 & 8 & X & B & 6 & 9 \\ 3 & 3 & & 8 & & 8 & 6 & 9 \end{pmatrix}$$

Fig. 9 Registration: using Levenstein algorithm is sometimes ambiguous. See text

between different alphanumerics (i.e. the substitutions) in the following way:

$$C_s = dist(C[i], B[j]) = \sum_{k=1}^{35} \left| C[i]_k - \frac{B[j]_k}{N_j} \right| \tag{5}$$

where $C[i]_k$ and $B[j]_k$ are the k^{st} elements of the classification result, representing the probability of the corresponding class, and N_j is the number of times the considered element of the buffer has been updated. Using our adapted distance gives a measurement that is more accurate because it uses the confidence of the OCR. For example, it is justified to preferably support a matching between a '8' and a 'B' than between a '8' and a 'X', because there is a larger ambiguity in the former case. Thus, the single association found by our algorithm in the case of Fig. 9 is $\begin{pmatrix} 3 & 3 & 6 & 8 & X & B & 6 & 9 \\ 3 & 3 & & 8 & & 8 & 6 & 9 \end{pmatrix}$.

5.2.2 Updating OCR during tracking

Once a correspondent $C[i]$ has been found for each buffer element $B[j]$, B is updated as follows: $B[j] = B[j] + C[i]$, where $C[i]$ is the 35-dimensional vector output by OCR. For example, Fig. 10a shows the buffer output for character 'B' in the example of the tracked vehicle which LP is '3368XB69'. When a vehicle outcomes from the video, the system returns for each $B[j]$ the class with maximal confidence. We can notice in Fig. 10a that the averaging has successfully detected 'B' as the maximum a posteriori class, and that all other classes have very low confidence, except '8'. Finally, only the elements of B that exceed a given confidence level C_l are returned as LP characters. This is illustrated in Fig. 10b, C_l being represented by a green line. Thus, wrongly extracted characters at some frames (here, '1' and '2' have been extracted in at least one frame) are filtered out, because they are not supported over time by our registration/update algorithm with a sufficient confidence. As we verify experimentally, this is a powerful way to improve extraction performances (see Sect. 7.2).

5.2.3 Final output of the system

We propose an approach for returning a set of probable LPs (instead of just the most probable), that is particularly useful in access control applications. Thus, we propose an algorithm dedicated to sample the a posteriori distribution of the combined kept buffer elements. The goal is to find a set of combinations whose probability is above a given percentage ρ_p of the best candidate probability (denoted $P_{\max}$). Formally expressed, considering a Buffer B with M characters,

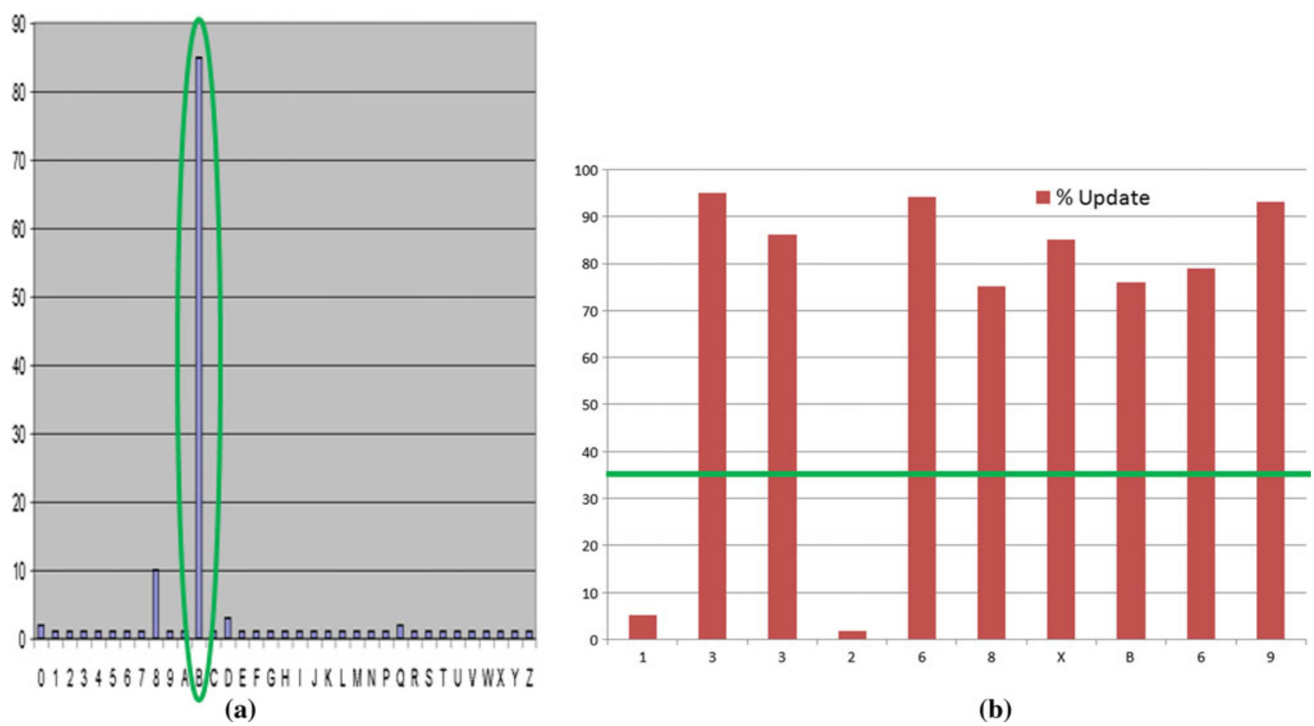

Fig. 10 Building an video-based LPR system: **a** buffer for 'B', **b** LP estimate: '-3368-XB69'

we want to find a sequence of index $\{k_1, \ldots, k_M\}$ whose probability P fulfills:

$$P\{k_1, \ldots, k_M\} = \sum_{j=1}^{M} B[j]_{k_j} > \rho_p * P_{\max} \tag{6}$$

To find the possible combinations that satisfy Eq. 6, we start by sorting the output vector $B[j]$ of each buffer element in decreasing order. At that stage, we have $P_{\max} = \sum_{j=1}^{M} B[j]_0$. We then compute the probability P_j by substituting $B[j]_0$ with $B[j]_1$, $j \in \{1; M\}$. We keep the substitution that leads to the maximum P_j, if $P_j > \rho_p * P_{\max}$. The process is then recursively applied.

6 Feedback among the different levels

In this section, we detail the cognitive loops that are used in our LPR system to improve its overall performance. They apply at many points of the proposed approach, e.g. LP tracking is an important example. These feedback steps are dedicated to overcome the lack of specificity of the low-level steps, that are unavoidable because of our national invariance requirement. Thus, at some critical stages of the algorithm, it analyzes its level of confidence, possibly leading to suspect the lower level decisions. Using vertical cooperations among the different levels corresponds to a form of reasoning that is close to the mechanisms of human perception. In our approach, five increasing logical levels are proposed, from the LP detection to the temporal averaging. Therefore, we claim that these top-down verifications loops are completely adapted to the LPR context, and are important aspects toward the design of intelligent and robust vision LPR systems that can deal with various LP national types.

6.1 Cognitive loops from extraction

The Connected Component Labeling (CCL) strategy for character extraction, described in Sect. 3.3, is accurate. However, it remains sensitive to noise, as a single misclassified pixel may lead to merge several characters, or split a single alphanumeric symbol into several components. To overcome this shortcoming, we analyze the aspect ratio bounding box $\rho = \frac{W}{H}$ of each extracted connected component. Depending on this analysis, either the segmentation is considered to be successful, or we use another strategy (e.g. projection histogram) that makes the segmentation more robust. We model the PDF (Probability Density Function) of ρ for a single character, $P(\rho/c(1))$, and the PDF of the distance between two

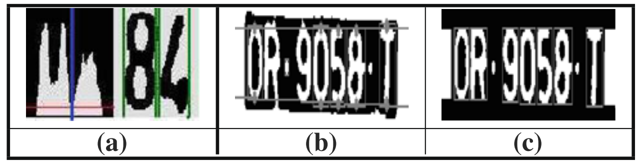

Fig. 11 Cognitive loops make single character segmentation more efficient

successive characters $P(d)$ with normal distributions:

$$\begin{aligned} P(\rho/c(1)) &= \frac{1}{\sqrt{2\Pi}\,\sigma_{\rho_S}} e^{-\frac{(\rho-\mu_{\rho_S})^2}{2\,\sigma_{\rho_S}^2}} = \eta(\rho, \mu_{\rho_S}, \sigma_{\rho_S}) \\ P(d) &= \frac{1}{\sqrt{2\Pi}\,\sigma_d} e^{-\frac{(d-\mu_d)^2}{2\,\sigma_d^2}} = \eta(d, \mu_d, \sigma_d) \end{aligned} \tag{7}$$

We experimentally validate the normality assumption of Eq. 7 in our database (see Sect. 7). Thus, the PDF of ρ for k characters can be expressed by $P(\rho/c(k)) = \eta(\rho, \mu_{\rho_k}, \sigma_{\rho_k})$, with $\mu_{\rho_k} = k \times \mu_{\rho_S} + (k-1) \times \mu_d$ and $\sigma_{\rho_k} = k \times \sigma_{\rho_S} + (k-1) \times \sigma_d$. The parameters of the distributions have been learnt with a standard EM technique [16]. Regarding character extraction, we estimate the number of characters in each connected component by evaluating the likelihood of $P(c(k)/\rho)$, $k \in \{1; N\}$,[1] and identify $\{k_p\}$, such that $P(c(k_p)/\rho) > T$, where T is a given threshold. If only $k_p = 1$ outputs, the component labeling is considered successful. Otherwise, we use the Histogram Projection (HP) technique to improve the robustness of the method. Thus, we compute a histogram that accumulates the vertical foreground pixels of its bounding box. The histogram is then processed to isolate the characters inside the blob, using $\{k_p\}$ as a *prior* to improve the segmentation (Fig. 11a). In the proposed cognitive loop, we estimate the CCL confidence level, and automatically detect its failure to use HP in that case. We experimentally validate (Sect. 7) that this feedback step significantly increases the extraction performances, and is more efficient than combining CCL and PH "horizontally".

Figure 11 illustrates another loop performed after the CCL extraction. Thus, we set all pixels of the image that are "outside" the region of the characters to background, defined by the two regression lines used for estimating the metric rectification (see Fig. 4). This is illustrated in Fig. 11b. From this, we start the CCL again, which makes it possible to segment characters which were ignored in the first pass of the algorithm, for example because it merged with some part of the radiator grille (see Fig. 11c).

Beyond individual character segmentation, Fig. 12 presents feedback steps that operate in a more global way. Thus, we evaluate the confidence level of the extraction step

[1] Note that we learn the case where a single character is split, $P(k < 1/\rho)$, so that our approach can manage this case.

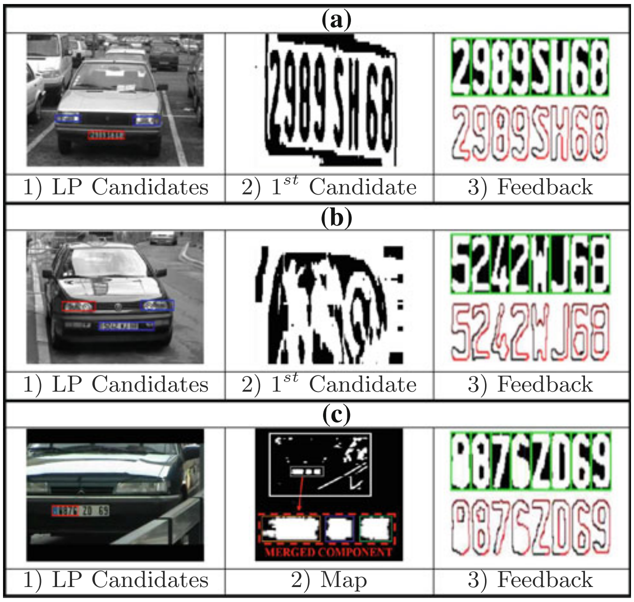

Fig. 12 Feedback on extraction: **a** from extraction to extraction: LP "Inverting", **b** from extraction to detection: next LP candidates, **c** from extraction to detection: fragmented LPs

depending on the number N of extracted characters. If N is highly too small or too large ($N \notin [N_{T_1}, N_{T_2}]$), the system loops as follows. First, the binary image of the plate is inverted. This aim at managing plates where the symbols are bright on a dark background (see Fig. 12a). If the extraction still fails, the area of the LP is supposed be wrongly identified. The algorithm comes back to the detection step, by analyzing the next LP candidate. This is illustrated in Fig. 12b. Figure 12b1 shows the initial image, with the first LP C_1 framed in red, the following candidates C_i $(i > 1)$ framed in blue. The LP does not correspond to C_1 because a light of the car gets a better score with respect to Eq. 1. The binary image of C_1, shown in Fig. 12b2, does not lead to any character extracted even after the binarization swapping. However, analyzing the second candidate C_2, corresponding to the LP, makes it possible to properly extract every character (Fig. 12b3). Wrongly identifying the LP with our non-country specific detection approach is actually unavoidable. Our detector happily fires in every area with a strong "vertical edges density", leading to false positives (e.g. lights, radiator grilles, building, trees, etc.). Finally, Fig. 12c illustrates a last cognitive loop, dedicated to handle cases where only a sub-part of the LP has been segmented. Thus, if $N \in [N_{T_1}, N_{T_2}]$, but does not lead to a high confidence level, we look for areas in the gradient accumulation image that are "close" to the current analyzed candidate. If such a candidate exists, we then compute the score of Eq. 1 of the resulted merged region. If this score exceeds the single candidate's score, the merge is validated. Then the algorithm starts back from the binarization step.

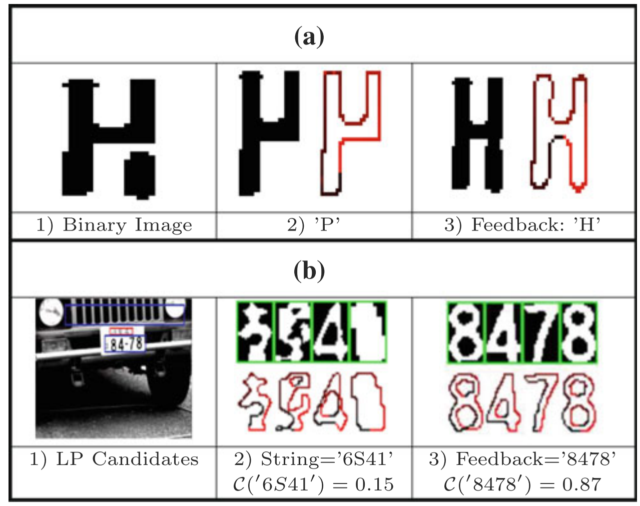

Fig. 13 Feedback on recognition: **a** from recognition to recognition: fragmented characters, **b** from recognition to detection: global recognition evaluation

6.2 Cognitive loops from recognition

Regarding OCR, the relevance of the algorithm is evaluated at two different levels, illustrated in Fig. 13. Let us consider the string C recognized by the system, constituted of N characters $C[i]$. First, we estimate the confidence level of the OCR for each character identified $\mathcal{C}(C[i])$:

$$\begin{aligned}\mathcal{C}(C[i]) &= w_r C[i]_0 + w_d \frac{C[i]_0}{\sum_{j=1}^{T} C[i]_j} \\ &= w_r \mathcal{C}_r(C[i]) + w_d \mathcal{C}_d(C[i])\end{aligned} \tag{8}$$

where $C[i]_0$ is the output class with the maximum a posteriori probability, and $C[i]_j$, $j \in \{1; T\}$ correspond to the T classes ranked below $C[i]_0$. $\mathcal{C}(C[i])$ is thus decomposed into $\mathcal{C}_r(C[i])$, that scores the recognition performance (i.e. how well the character is identified), and $\mathcal{C}_d(C[i])$, that scores the discriminability performance (i.e. how well the character is discriminant with respect to its T competitors). If $\mathcal{C}(C[i])$ is not enough, we try to merge other connected components with $C[i]$, and evaluate $\mathcal{C}(C[i])$ again. For example, the binary image of the 'H' shown in Fig. 13a1, is recognized as a 'P' (Fig. 13a2).[2] After the merge, the character is then recognized as a 'H', with a good confidence level (Fig. 13a3).

In addition, the performance of the recognition is evaluated at a global level. Thus, we define the confidence level $\mathcal{CL}(C)$ of the string C as follows:

$$\mathcal{CL}(C) = \frac{1}{N} \sum_{i=1}^{N} \mathcal{C}_r(C[i]) \tag{9}$$

[2] Indeed, it is matched to a 'P' that is learnt with a topological error, see Sect. 4.

Again, $\mathcal{CL}(C)$ is matched against a given threshold to validate the recognition. For example, the LP detected as shown in Fig. 13b1 is not estimated to be valid, because the output string ('6S41') (see Fig. 13b2) is not confident enough. Indeed, except the '4', other symbols belong to the Japanese alphabet, and each partial score $\mathcal{C}_r(C[i])$ is thus very low. A feedback loop is then activated, and the string '8478' is recognized with a high $\mathcal{CL}(C)$, as illustrated in Fig. 13b3.

6.2.1 LP format estimation

Finally, an important cognitive loop consists in estimating the LP format, and using this information to improve OCR accuracy. However, we want to stress that LP format estimation is only an optional step in our method, although this strategy is the heart of other approaches for multinational LPR, e.g. Refs. [19,40]. In our approach, we rely on cognitive loops to perform multinational recognition, and LP format estimation is only one of the various cognitive loops. In the experiments, the majority of the evaluation has been carried out without format estimation.

To perform format estimation, we firstly cluster the different extracted characters into groups, based on spatial proximity in the image. Then, we assign to each group a given label belonging to a *prior* alphabet $\mathcal{A} = \{a\}$. As each group G_i is composed of a sequence of symbols S_i, where S_i is the OCR output, we choose to model G_i with a Markov Chain, and use training data to learn the parameters with a EM algorithm. During alphabet recognition, we maximize for each group G_i the a posteriori probabilities $P(a|G_i)$, $a \in \mathcal{A}$. Maximizing $P(a|G_i)$ can be solved efficiently using dynamic programming. Once G_i has been assigned its most likely label, the LP is modeled as a string of labels. Finding the most likely format is then carried out by minimizing the Edit Distance between the actual string of labels making up the LP, and the different *prior* LP format models.

7 Experimental results

Some images of the sample datasets used for validation are shown in Fig. 14. The static database contains two subsets. Subset S1 has been downloaded from http://x.htsol.com/. It contains 365 images composed of vehicles from 15 different nationalities. Subset S2 contains 843 images captured at a toll highway, mainly composed of European LPs. It presents extremely difficult examples with strong variations in distance between the camera and the LP, as well as strong variations in the LP orientation. Moreover, difficult elements such as vehicle's light, specular effects with reflects, low contrast images and fog have to be taken into account. The video dataset is composed of 6 different subsets. The authors have manually collected it. In all the examples, the vehicles never stop, which makes the tracking task challenging. The subsets S1, S3 and S6 are composed of 51, 89 and 76 English and French fast moving vehicles, respectively. There are front and rear views, and one have to face several LPs simultaneously present, with vehicles overtaking each other. The subset S2 contains 124 vehicles at a car park exit, slowly moving in night conditions. In the subset S4, there are mainly French and German cars, with a total number of 144 vehicles, moving moderately in opposite directions. Finally, the subset S5 is composed of 41 Dutch cars moderately moving. The video is acquired in raining acquisition conditions.

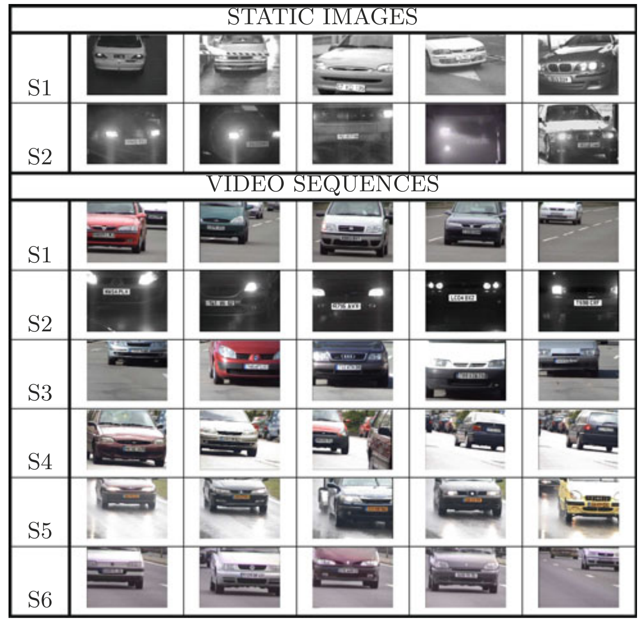

Fig. 14 Sample datasets used in the evaluation

We present here the performance of the different steps of the algorithm, as in standard evaluations. However, we emphasize the validation on real data of our three main contributions: the use of cognitive loops, the hybrid statistical/structural OCR, and the temporal averaging.

7.1 Validation on static images

7.1.1 Performances evaluation on plate detection

To evaluate the detection performances, we manually label each LP bounding box in the dataset, and we define a positive (negative) as an image with (without) LP. Let us denote N_n and N_p as the number of positives and negatives, respectively, in a subset composed of N_t images. True positives are identified by a "sufficient" overlap, which is observed between the LP bounding box and the manually labeled one, using our tracking criterion (Eq. 4). With N_{tp} true positives and N_{fn} false negatives identified in the subset, we can thus compute the detection rate D_r and the false detection rate

Table 1 LP Detection: cognitive loops sensitively increase performances, see text. *D* detection, *T* truth

D	(a) Bottom-Up				(b) Feedback			
	S1		S2		S1		S2	
	N_p	N_n	N_p	N_n	N_p	N_n	N_p	N_n
T								
N_p	339	269	622	123	358	7	702	43
N_n	0	0	16	82	0	0	26	72
D_r (%)	92.9		83.5		98.1		94.2	
FD_r (%)	0.0		16.3		0.0		26.5	

Class	Number of errors	Number of samples	Error rate
$0^{\square}$	25	248	10.08
1	0	241	0.60
$2^{\triangle}$	4	263	1.58
3	3	290	1.03
4	3	175	1.71
5	0	269	0.00
6	6	381	1.56
7	1	261	0.38
$8^{\diamond}$	98	410	23.90
9	1	335	0.30
A	1	191	0.52
$B^{\diamond}$	35	276	12.68
C	0	95	0.00
$D^{\square}$	15	156	9.62
E	0	63	0.00
F	0	55	0.00
G	4	140	2.86
H	0	76	0.00
I	1	56	1.79
J	0	78	0.00

Class	Number of errors	Number of samples	Error rate
K	0	134	0.00
L	0	85	0.00
M	0	103	0.00
N	0	88	0.00
P	0	183	0.00
$Q^{\square}$	5	59	8.47
R	0	158	0.00
S	0	131	0.00
T	0	53	0.00
U	0	53	0.00
$V^{\diamond}$	6	76	7.89
W	0	72	0.00
X	0	78	0.00
$Y^{\diamond}$	0	61	0.00
$Z^{\triangle}$	0	123	0.00
HNN Performances	208	5592	3.72
HNN + Structural	162	5592	2.89
Flat Neural Net + Pixels	862	5592	15.41
Flat Neural Net + FD	402	5592	7.2

Fig. 15 Performances of our own OCR algorithm, on testing. Our HNN overpasses flat Neural Nets with pixels and FDs input. Combining HNN with structural classifier significantly improves similar character recognition rate. See text

FD_r in the following way:

$$D_r = 100 \times \frac{N_{\mathrm{tp}}}{N_p}$$
$$FD_r = 100 \times \frac{N_{\mathrm{fn}}}{N_n} \tag{10}$$

Table 1 presents the LP detection performance with our proposed approach. Table 1a) shows the evaluation without any feedback, i.e. when only using the bottom-up processing detailed in Sect. 3.1. The recognition rate for S1 and S2 reaches 92.9 and 82.0%, respectively. If the performances for S1 are already very good, the system has more difficulties to face illumination variations, lower resolution images, background clutter etc. Table 1b presents the evaluation results with the cognitive loops introduced in Sect. 6. It is noteworthy that the detection accuracy is highly increased for both S1 and S2. The recognition rate reaches 98.1% for the subset 1, and 94.2% for the subset 2. The remaining non-detected Lps in S1 come from LPs merging with surrounding areas with high vertical gradient density. In S2, they are mostly due to extreme low contrasted images (because of the fog), making the vertical gradient detection fail. Finally, we can notice that FD_r is not extremely low for subset 2. We can, however, tolerate a limited precision in our application, since the higher level processing can easily filter out false positives.

7.1.2 Performances evaluation on plate recognition

First, we evaluate our own OCR algorithm, i.e. we only estimate the OCR performances irrespective of other bottom-up steps.

Resulting tests are presented in Fig. 15. As we can see, the statistical classifier, e.g. HNN, outputs an overall recognition rate of 96.3%. As a comparison, we evaluate the performances using a flat neural network. The recognition rates using image pixels as input is dramatically smaller, only reaching 84.6%. This validates the fact that this feature is not relevant for our classification purpose. Using FDs as input makes the recognition rate reach 92.8%, which proves that FDs constitute a robust image feature. However, the flat classifier still has difficulties to learn characters with topological variations, contrarily to our HNN. This justifies the validity of the HNN to accurately discriminate the majority of the classes. However, Fig. 15 illustrates as well the limitation of FDs to discriminate similar pattern. Indeed, a large disparity among the different classes exists, the error rate for the similar characters ('8'/'B', '2'/'Z', '0'/'D'/'Q', and 'V'/'Y') being extremely higher than for the other classes. Indeed, these classes only differ locally and are only distinguishable by salient features such as right angles that are lost after the lowest frequency terms extraction. If we do not take into account these characters, the recognition rate with the HNN reaches 99.5%. For similar pattern, combining the HNN with the structural classifier significantly increases performances, as illustrated in Fig. 15. Our combined approach shows a global error rate about 2.89%, instead of more 3.72% for the HNN based classifier. For the most difficult class, i.e. '8'/'B', we get an improvement of 13%, from 80% to 93%. Then, there is an improvement of 0.6% for '2'/'Z' images, 1% for 'V'/'Y' images, and 2% for '0'/'D' images. In Table 2, we depict the best rates of hybrid classification for various bases of similar images.

In addition, we evaluate the recognition performances including the low-level steps after LP detection (i.e. binarization, extraction and OCR). For a given subset containing M LPs, let us denote T_p^i the string of N_i characters corresponding to the ist LP, and D_p^i the string detected with our algorithm. We then compute the Levenstein distance $L_d^i(T_p^i, D_p^i)$ [27] as a metric for estimating the quality of the OCR. The OCR rate R_c is then defined in the following way:

$$R_c = \sum_{i=1}^{M} \frac{N_i - L_d^i(T_p^i, D_p^i)}{N_i} \tag{11}$$

The LP recognition rate R_p corresponds to a proper identification of every character of the true vehicle T_p^i, leading to the following estimation for R_p:

$$\begin{aligned} R_p &= \frac{1}{M} \cdot \sum_{i=1}^{M} B^i(T_p, D_p) \\ B^i(T_p, D_p) &= \begin{cases} 1 & \text{if } L_d^i(T_p, D_p) = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned} \tag{12}$$

The recognition performances based on R_c and R_p are presented in Fig. 16. The blue and green curves illustrate R_c and R_p, respectively. In addition, the red curve shows the overall performances including LP detection. To point out our main contributions, we present the performances at incremental progress steps. First, we evaluate the system only using bottom-up steps, and only using the statistical approach for OCR (step 1). Then, we present the relative gain brought out by using cognitive loops (step 2) and the combined OCR approach including structural recognition (step 3). For both subsets S1 and S2, we can notice that cognitive loops sensitively improve performances (33 and 55% gap for R_p, respectively). Regarding combined OCR approach, the relative gain is large for S2 (15% gap for R_p), as it is more limited for S2 (1.5% gap for R_p). Indeed, S1 mainly holds high quality images, since S2 contains mainly low-resolution and damaged images.

For S2, the overall recognition rate including detection reaches 69% at step 3, which may seem not very efficient. However, we would like to stress that this subset holds very damaged images and various LP types. When processing static images, our multi-national system that prevents the use of specific syntactic rules is still limited by the extraction step. We show in Sect. 7.2 that using the temporal aspect of the video sequence can efficiently filter out these errors. Finally, we present in step 4 two possibilities for improving recognition with static images. For S2, we compute a set of probable LPs, as proposed in Sect. 5.2.3), and validate the recognition if one of them corresponds to the true LP. As we can see in Fig. 16b), the gain for R_p is 11%. This solution is however only applicable in the access control context. Another attempt for improving the performances while maintaining a multi-national recognition consists in automatically detecting the LP format, as explained in Sect. 6.2.1, and using this estimation to make the recognition much more efficient. In our experiments, we concentrate in recognizing the following European LPs: English, Spanish, French, Dutch, German and Italian. Thus, we define the following alphabet for each group of characters $\mathcal{A} = \{D, L, E, I\}$. D, and L correspond to groups that only contain digits or letters, respectively. E is a specific group of some English plates, composed of letter 1 followed by three digits. I is a specific group of some Italian plates composed of more than 5 characters. Each group G_i is thus assigned the most probable label of the alphabet, modeling G_i with a Markov chain. It leads to form a string

Table 2 Recognition rates, depending on the method used and the set of images treated

Images	Neural network (%)	Hybrid classification (%)
'8'/'B'	80.61	93.64
'2'/'Z'	98.96	99.63
'V'/'Y'	95.62	96.63
'O'/'D'	90.10	92.19

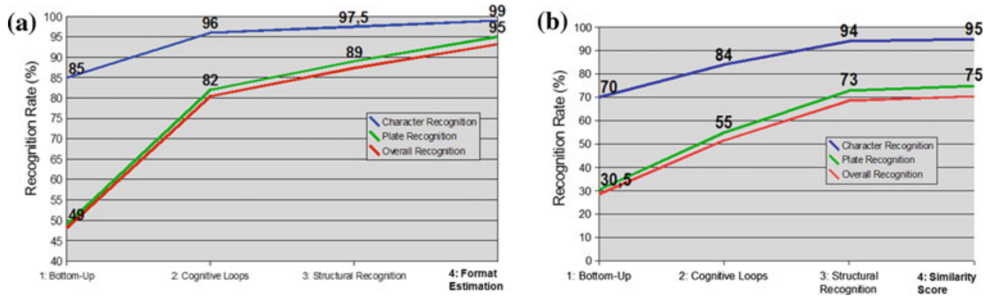

Fig. 16 Performance evaluation on recognition: **a** subset 1 and **b** subset 2

Table 3 LP national format estimation

	Fr	Sp	Ne	UK	Ge	It	Ov
NIR	95.2	96.3	100	97.1	95.5	98	97.5
NPRR	100	96.8	100	96.3	99	96.4	98

NIR Nationality Identification Rate (%), *NPRR* National Plate Recognition Rate (%), *Fr* France, *Sp* Spain, *Ne* Netherland, *UK* United Kingdom, *Ge* Germany, *It* Italy, *Ov* overall

of labels. This string is then matched, by computing the Edit Distance, against the seven followings models: DLD (French plates), DL (Spanish plates), EL (English plates), LI (Italian plates), DLL (Dutch plates), LLD and LDL. As LLD and LDL are ambiguous, we use the intended number of characters within each group to make the final decision. Table 3 presents the Nationality Identification Rate, reaching 97.5% for the overall dataset. Once LP format is identified, the LP recognition rate is excellent, as shown in Table 3: we get 98% in average. When applied to S1 subset, we get a gain of 6%, reaching 95% (see Fig. 16a). Indeed, the format identification can take advantage of syntactic rules to significantly improve extraction (by filtering out false positives and detecting missing characters by context) and recognition (there is a strong digit/letter *prior*).

7.2 Validation on image sequences

7.2.1 Evaluation setup

The temporal integration described in Sect. 5 detects single vehicles, tracks them, identifies their entries and outcomes from the field of view, and outputs a final recognition for the LP. The system thus returns a sequence of N strings D_p^i, $i \in \{1; N\}$, with start and end time points that we denote S_d^i and E_d^i, respectively. Let us denote the detected temporal bounding box $D_{t_{BB}}^i$ of the i^{st} tracked LP, identifying the area in the image where the LP was during the analysis. To evaluate the performances, we manually label all the videos with that standard, leading to store a set of M true LPs T_p^i, $i \in \{1; M\}$, with start, end points and temporal bounding boxes S_t^i, E_t^i and $T_{t_{BB}}^i$, respectively. We thus define a matching score $MS(D_p^i, T_p^j)$ between each pair of detected and true LP as follows:

$$MS(D_p^i, T_p^j) = w_t \times O_t(D_p^i, T_p^j) + w_g \times O_g(D_{t_{BB}}^i, T_{t_{BB}}^j) \quad (13)$$

where $O_t(D_p^i, T_p^j)$ corresponds to the temporal overlap between D_p^i and T_p^j, and $O_g(D_{t_{BB}}^i, T_{t_{BB}}^j)$ is the image overlap between the detected and true temporal bounding boxes. We compute a binary matrix of size $M \times N$ that contains all the possible correspondences between the detected and labeled LPs, by matching $MS(D_p^i, T_p^j)$ to a given threshold MS_T. In case of multiple matches for a given LP, we only keep the best correspondent by minimizing the Levenstein distances among the set of ambiguities. This leads to associate to each labeled LP at most one detected LP, and vice versa. The detection rate D_r and the false positive rate FD_r can then be estimated as defined in Eq. 10. Identically, the recognition rates for the characters and the LPs can be evaluated as defined in Eqs. 11 and 12 for the static part of the evaluation.

7.2.2 Results

Table 4 presents the 6 subsets evaluation. As shown in the first row, the LP detection rate for the overall subsets is very good, reaching 98.29%. For the 529 vehicles, our system only fails in detecting nine License Plates. These mis-detections were due to bad quality images resulting from difficult acquisition conditions and to tracking failures. Improving tracking with finer image features is a direction for future works (Sect. 8). Regarding LP recognition, the global rate is about 95.16%. This illustrates the efficiency of the temporal averaging, compared to the recognition rates on static images (Fig. 16). To point out the improvement due to the temporal averaging, we evaluate the plate recognition rate by only considering the static images independently, being around 80%. This score may seem to be low, and is actually a bit biased. Indeed, as the vehicles exit from the field of view, their license plate is only partially visible in the image, leading to an over-esti-

Table 4 License Plate Recognition performances on image sequences

	S1	S2	S3	S4	S5	S6	Ov
PDR (%)	96.1	99.2	100	97.9	95.1	98.7	98.3
PRR (%)	89.8	98.4	94.4	92.9	94.9	96.7	95.2
ORR (%)	86.3	97.6	94.3	91	90.2	97.4	93.5

Ov overall, *PDR* Plate Detection Rate, *PRR* Plate Recognition Rate, *ORR* overall recognition rate

Table 5 Processing time of the different steps of the algorithm

	Det	Bin	Ext	OCR	Av	Cog	Ov
Time (ms)	15	19	9	24	5	?	72

Det LP detection, *Bin* binarization, *Ext* character extraction, *Av* temporal averaging, *Cog* cognitive loops, *Ov* overall

mated error rate. However, it points out the capacity of the proposed temporal averaging algorithm to efficiently combine the different recognition results over time. Regarding extraction, the capacity to manage insertions and deletions between the different strings is an important aspect that sensitively improves the recognition efficiency. As explained in the static evaluation, an important part of the remaining errors still comes from the extraction step. This illustrates the efficiency of the registration algorithm (Sect. 5.2.1) and the update approach (Sect. 5.2.2) that only keep the most significant elements of the averaged buffer. Regarding recognition, the temporal update of the classifier output sensitively improves the recognition rate on individual characters, particularly for ambiguous classes. Indeed, the algorithm take advantage of high resolution images (e.g. when the vehicle is close to the camera), for which the recognition is more reliable.

7.3 Processing time

Table 5 gathers the complexity for each step of the algorithm. This experiment has been carried out on a Pentium IV at 2.66 GHz, with 512 MB of RAM. The software program has been built using Microsoft Visual studio 2005 (version 7.0). The performances are given for images with 4× CIF resolution (i.e. of size 640 × 480), and for not optimized code.

As we can see, there are three main demanding steps: LP detection (that requires a per-pixel processing), binarization (because we use a local approach for managing illumination variations), and OCR. Note that the processing time for OCR is given for a LP with eight characters, since the average time per character is around 3 ms. The bottom-up steps of our system can thus be processed at a pace of about 15 frames per second, which is near real-time. Processing time for cognitive loops is very difficult to estimate, as it depends on various factors: image content, step from which feedback is activated, number of loops, etc. With static images, time limitations are not so strong, and our algorithm always finishes in a reasonable time (i.e. there is no problem to process our database). When dealing with videos, however, we must in practice bound the computation time in order to make sure a sufficient number of frames is analyzed. In our experiments, we notice that we can analyze our image sequences with a small loss of accuracy at a pace of ten frames per second (i.e. with a maximal processing time of 25 ms for cognitive loops).

7.4 Comparison with existing systems

An exhaustive review of the existing methods up to 2005 has been proposed by Anagnostopoulos et al. [1]. We concentrate here on systems that have been validated in images or videos containing a significant number of vehicles (at least 1000). The comparison of different LPR systems is a difficult task, because of the lack of a common benchmark and the strong variability in the acquisition conditions (image resolutions, viewpoints, scales, etc.). We do not attempt here to evaluate the commercial LPR systems, because their technical background is strictly confidential, and moreover, their performance rates are often overestimated for promotional reasons. However, even for the non-commercial systems, huge differences in the application settings exist. Plus, in country-specific systems, the *prior* knowledge on LP format is so strong that a direct comparison with multinational approaches is irrelevant. Therefore, we propose to classify the different methods based on three criterions (see Table 6).

The left column is composed of country-specific systems, i.e. clearly dedicated to a specific type of national LP for the detection or the recognition. As we can notice, the performances are very impressive, and some of these methods were proposed more than 15 years ago. For example, the system proposed by Naito et al. [31] dedicated to recognize Japanese plates outputs a detection rate of 97% and an impressive plate recognition rate of 99.3%. However, the dataset is composed of high-resolution images where the plate occupies the whole image. Above all, we again want to stress that *prior* LP format makes LPR task easy. Indeed, the system aims at identifying four registration numbers (there are only ten output classes). In addition, the digit extraction is strongly constrained by the *prior* LP format: digits are supposed to be above Japanese symbols (representing district, classification number and prefix). The systems of Chang et al. and [6] addresses difficult situations: complex background, various environments (street, roadside, and parking lot), and different illumination, some of them with damaged LPs. The plate recognition rate reaches 95.6% on static images. However, their algorithm still use syntactical

Table 6 LPR system classification

Country-specific systems			Multi-national systems			Video-based systems		
References	Performances		References	Performances		References	Performances	
	D	R		D	R		D	R
[32]	78.75%	98.51%(ch)				[9]	Not reported	
[8]		91% (pl)	[35]		95.4% (ch)	[10]	100%	50% (pl)
[11]	98%	80% (pl)	[33]	S : 84.5% (ch)		[2]	99%	90%
[23]	100%	94.7% (pl)	[3]	98.7%	98% (ch)	[42]	Not reported	
[31]	97%	99.3% (pl)	[1]	96.5%	89.1% (pl)			
[6]	97.6%	95.6% (pl)	[40]	81.2%	85.2% (ch)			
[12]	98.8%	95.6% (pl)	[19]	95.9%	90% (pl)			

D detection rate, *R* recognition rate, *S* segmentation rate, *pl* plate, *ch* character

rules for the final recognition that highly improve the overall performances.

The second column of Table 6 shows the performances of approaches that attempt to manage a wide range of national LPs. Obviously, these methods are not effective for any country, but they are not strictly limited to a single LP format. Furthermore, they do not rely on a verification step using *prior* LP format to identify valid license numbers. For these multi-national approaches, the performance variation is here significant. Indeed, the recognition rate on characters only reaches 85.2% in [40] (i.e. a recognition rate of less than 40% on LPs with 6 characters). In [1], the overall recognition rate (detection and LP recognition) is 86%. In particular, they evaluate their performances on a subpart of our subset 1 (185 images on 365). They report an LP recognition rate of 89%, i.e. similar to our approach at step 3 (see Fig. 16). Note that using our national formal estimation, we can reach a much better recognition rate of 95%. In addition, they report that their algorithm has difficulties with similar characters, and they do not use any structural approach to overcome that shortcoming. The recent approach [19] is the most related to ours because of its strong multi-national purpose. Their main contribution is to parametrize their system for modeling different LP formats, and using the detected format for improving the recognition. They evaluate their approach on various LP types, with alphanumerics and other Chinese symbols. The overall recognition rate is 90%. Interestingly, they compare their approach with [1] when varying LP from one style to another, and report that their performances remain steady while a drop of about 9% is observed when using the system in [1]. However, as pointed out in the introduction, although the approach in [19] has the same application purpose than our system (i.e. being robust to LP format variations), their methodology significantly differs: since they parametrize the low-level steps for managing various nationalities, we rather rely on using cognitive loops for improving lower level performances.

Finally, the third column of Table 6 presents the performances of approaches that explicitly incorporate temporal information to perform the recognition. Both approaches [2, 10] perform LP detection using a Viola-Jones classifier, followed by a tracking scheme. The reported performances are very good, i.e. more than 99%. The plate recognition rate is however not good (50%) in [10]. The system makes use of a super-resolution technique to build a sub-sampled image and perform OCR on it. One can note that the LPR constitutes only a subpart of the authors work for recognizing cars. The OCR is performed with a simple template matching technique, probably explaining the low recognition rate. Moreover, the authors report their specific difficulties to identify similar characters. In [2], the performances on LP recognition reach 85 and 95% on the two subsets used. However, the approach is country-specific since it uses a syntax number checking technique as a post-processing step. Similar to our experiments, the recognition rate appeared to be significantly improved by combining the OCR results during the tracking process. However, they use a simple a simple majority voting technique to perform the temporal averaging. This strategy is not robust to insertion or deletion of characters over time. We can guess that it is properly performed only because of the country-specific context, i.e. because the known LP structure helps the registration step. Contrarily, our probabilistic edit distance approach can manage much more general settings.

8 Conclusion and future works

In this paper, we propose a video-based LPR system that strives for not being country-specific, making it applicable in realistic settings. To this purpose, using cognitive loops proves to be an efficient way to improve detection and recognition performances. In addition, the hybrid statistical/structural method provides an extremely efficient OCR, that does not need specific syntactic rules to properly identify LPs.

Finally, we propose a robust approach approach based on the probabilistic edit distance for averaging OCR results over time, that can take advantage of high quality images to improve the overall recognition.

Regarding future works, we are investigating how to generalize our approach to other LP national type classes, by managing multiple row LPs and various alphabets. Moreover, tracking could be improved, by using finer image features. Finally, beyond character recognition, building visual models of vehicles is another promising direction to recognize them more accurately.

Acknowledgments This work has been supported by the SAS Foxstream.

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